

Manjunath Bestha

Senior Research Fellow

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Personal & Educational Details

DOB : **1996-Apr-01.**

Nationality : **Indian.**

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2022 – Sep 2026 **Ph.D. — Astronomical Instrumentation**, Indian Institute of Astrophysics, Bengaluru & Calcutta University, Kolkata, India.

Title: **"Exoplanet Transmission Spectroscopy: Instrumentation and Observations"**
Supervisors: **Prof. Sivarani Thirupathi**

Astronomical Instrumentation, Transmission Spectroscopy, Spectroscopy Simulations, Machine Learning

2019 – 2021 : **Master of Technology — Astronomical Instrumentation**, Indian Institute of Astrophysics, Bengaluru & Calcutta University, Kolkata, India.

2013–2017 : **Bachelor of Technology — Mechanical Engineering**, Jawaharlal Nehru Technological University, Anantapur, Andhra Pradesh, India.

Research Experience

2023–present **Senior Research Fellow**, Indian Institute of Astrophysics, Bangalore.

Observations and development of data analysis pipelines of exoplanet atmospheres using both low- and high-resolution ground-based transmission spectroscopy with data obtained from the 2m Himalayan Chandra Telescope (HCT).

Exoplanet transit modeling, including Rossiter–McLaughlin and centre-to-limb variation (RM/CLV) effects

Identification of sources of observational systematics and modeling/mitigation in photometric and spectroscopic datasets using one-dimensional and two-dimensional Gaussian Process frameworks.

End-to-end simulations of multi-object transmission spectroscopy observations for next-generation multi-object high-resolution spectrographs.

Measurement and characterization of atmospheric dispersion using the Hanle Faint Object Spectrograph and Camera (HFOSC) mounted on the Himalayan Chandra Telescope.

2021–2023 **Junior Research Fellow**, Indian Institute of Astrophysics, Bangalore.

Optical design of a broadband atmospheric dispersion corrector and fore-optics using ZEMAX.

Conceptual mechanical design of a four-degree-of-freedom (4-DoF) fiber positioning system.

Development of a collision-avoidance algorithm for a moderately multiplexed 4-DoF fiber positioning system.

Throughput analysis and development of an Exposure Time Calculator for the Thirty Meter Telescope High Resolution Optical Spectrograph (TMT-HROS).

Optical design analysis of the Thirty Meter Telescope High Resolution Optical Spectrograph (TMT-HROS).

Optical performance analysis of fiber-based spectrographs.

Design and analysis of microlens systems for fiber-based spectrographs.

Research Projects

- 1 **Ground-Based Spectroscopic Observations of Exoplanet Atmospheres with HCT.**
Conducted observational studies of exoplanet atmospheres using spectroscopic data obtained from the 2-m Himalayan Chandra Telescope (HCT). This work includes high-resolution transmission spectroscopy of ultra-hot Jupiters such as KELT-9 b and investigations of instrumental and wavelength-dependent systematics affecting multi-object transmission spectroscopy. The goal is to detect atmospheric species and improve the reliability of ground-based atmospheric characterization.
- 2 **Instrumentation Development for the Thirty Meter Telescope (TMT).**
Contributed to instrumentation studies for future Extremely Large Telescopes, particularly the Thirty Meter Telescope (TMT). This work includes the design and analysis of a broadband Atmospheric Dispersion Corrector (ADC) for HROS-TMT, development of concepts for a multi-functional fiber positioning system, and investigations of multi-object spectroscopic approaches for exoplanet atmosphere studies. These studies aim to improve observational efficiency and precision spectroscopy capabilities for next-generation telescopes.
- 3 **Simulations and Performance Studies for Spectroscopic Observations.**
Developed simulations and performance analyses for astronomical spectroscopic instruments and observational techniques. This includes multi object spectroscopy simulations for the TMT High Resolution Optical Spectrograph (HROS) and atmospheric dispersion measurements at the Hanle site, and studies of wavelength-correlated systematics in transmission spectroscopy using statistical and Gaussian Process-based approaches.

Research Interests

- 1 **Exoplanet Studies.**
 - Exoplanet atmospheric characterization using low- and high-resolution transmission spectroscopy
 - Detection of atomic and molecular species in exoplanet atmospheres using ground-based spectroscopy
 - Development of multi-object spectroscopic techniques for exoplanet atmosphere studies
 - Modeling and mitigation of observational systematics in transmission spectroscopy and photometry using AI/ML
- 2 **Astronomical Instrumentation.**
 - Optical design optimization and performance analysis of telescopes, spectrographs, and their subsystems
 - End-to-end optical design simulations
 - Designing optimal mechanical concepts for astronomical instruments
 - Instrument design requirements and key aspects of spectrographs
 - Data pre-processing, reduction, and pipeline development for low-, high-resolution, and multi-object spectra
 - Optical laboratory experiments, and observatory site and instrument characterization through the design of observational strategies and analysis of observational data

Tools & Skills

- 1 **Transmission spectroscopy.**
 - Experience in exoplanet atmospheric observations, data reduction, analysis, and synthetic model fitting
 - Experience in preprocessing fiber-fed, multi-object, slitless, and long-slit spectroscopic data
 - Experience using JAX, george, LUAS, emcee, PyLightcurve, PyTransit, and petitRADTRANS
- 2 **Other Tools & Skills.**
 - End-to-end simulation and ray tracing using Zemax and Python
 - Developed semi-automated codes for reducing fiber-fed high-resolution and slit, slitless low-resolution spectra
 - Data handling and data reduction tools in Python
 - High and low-resolution spectroscopic data reduction and photometry using PyRAF
 - Experience of 40+ complete night observations at HCT and VBT

Publications

Manuscripts in Preparation

- 2025 **Manjunath Bestha**, Unni A., Sivarani T., Surya A., Parvathy M., Divakar D., “**Probing the Atmosphere of KELT-9 b with High-Resolution Spectroscopy from a 2-m Himalayan Chandra Telescope**”.
- 2025 **Manjunath Bestha**, Unni A., Sivarani T., Surya A., Parvathy M., Divakar D., “**An Observational Study of Systematics Affecting Low-Resolution Multi-Object Transmission Spectroscopy**”.

Journal Articles

- 2025 **Manjunath Bestha**, Sivarani T., Wehbe B., Hasan A., Chandra B. P., Divakar D. K., Unni A., Menon P., Surya A., Saraf P., “**A New Broadband Atmospheric Dispersion Corrector for HROS-TMT**”, RASTI (2025).
<https://academic.oup.com/rasti/article/doi/10.1093/rasti/rzaf057/8327618>
- 2025 Unni A., Oza A. V., Hoeijmakers H. J., Seidel J. V., Sivarani T., Schmidt C. A., Kesseli A. Y., de Kleer K., Baker A. D., Gebek A., Meyer zu Westram M., Fisher C., Sallum S., **Manjunath Bestha**, Bello-Arufe A., “**Doppler Shifted Transient Sodium Detection by KECK/HIRES**”, arXiv:2504.03974.
<https://doi.org/10.48550/arXiv.2504.03974>

Conference Proceedings

- 2025 **Manjunath Bestha**, Unni A., Sivarani T., Menon S., Parvathy M., Surya A., Saraf P., Divakar D., Manickavasaham L., “**An Initial Assessment of the Hanle Echelle Spectrograph for Exoplanet Atmosphere Studies**”, SPIE (2025).
<https://doi.org/10.1117/12.3064331>
- 2025 **Manjunath Bestha**, Athira Unni, T. Sivarani, Dhanush S. R., Lokesh Manickavasaham, Parvathy M., Devika K. Divakar, Arun Surya, “**A Multi-Object Approach for Studying Exoplanet Atmospheres Using High-Resolution Spectrographs**”, SPIE (2025).
<https://doi.org/10.1117/12.3064406>
- 2025 **Manjunath Bestha**, Sivarani T., Unni A., Parvathy M., Divakar D., “**Atmospheric Dispersion Measurement at Hanle Site**”, Springer (2025).
<https://doi.org/10.48550/arXiv.2509.17300>
- 2025 Manickavasaham L., **Manjunath Bestha**, Sivarani T., Surya A., Unni A., “**Addressing Wavelength-Related Systematics in Exoplanet Transmission Spectroscopy: A 2D Gaussian Process Approach**”, arXiv:2510.20423.
<https://doi.org/10.48550/arXiv.2510.20423>
- 2025 Parvathy Menon, Sivarani T., Sriram S., **Manjunath Bestha**, Devika K. Divakar, S. P. Rajaguru, Arun Surya, “**High-Resolution Echelle Spectroscopy for Solar System Planets: A Planet-as-Point-Source Analogy**”, arXiv:2510.20559.
<https://doi.org/10.48550/arXiv.2510.20559>
- 2024 **Manjunath Bestha**, Sivarani T., Surya A., Yadav S., Unni A., Parvathy M., Divakar D., Sriram S., Prakash A., Hasan A., “**A Multi-Functional Fiber Positioning System for Extremely Large Telescopes**”, SPIE (2024).
<https://doi.org/10.1117/12.3019098>
- 2024 Devika K. Divakar, Thirupathi Sivarani, Ramya Sethuram, Jason Fucik, Chuck Steidel, **Manjunath Bestha**, Prasanna Deshmukh, “**Slitless Spectroscopy Simulations of Milky Way Dwarf Satellite Galaxies Using TMT-WFOS**”, Proc. SPIE 13099 (2024).
<https://doi.org/10.1117/12.3019070>

- 2023 **Manjunath Bestha**, Hasan A., Divakar D., Surya A., Sivarani T., Sriram S., Prakash A., Parvathy M., Yadav S., “**Atmospheric Dispersion Corrector for a Multi-Object Spectroscopic Mode of HROS-TMT**”, SPIE (2023).
<https://doi.org/10.1117/12.2677192>

Awards, Honors & Academic Distinctions

- 2025 **Early Career Researcher Award**, *International Conference on Space for Sustainability: Science, Technology, Education, and Policy (S²-STEP 2025) alongside the 6th Indian Planetary Science Conference (IPSC 2025)*, Indian Institute of Technology (IIT) Roorkee, India..
- 2025 **Best Poster Award**, *Astronomical Society of India Meeting on Sun, Solar System, Exoplanets & Astrobiology*, National Institute of Technology (NIT) Rourkela, India..
- 2024 **Early Career Researcher Award**, *International Conference on Planets, Exoplanets, and Habitability (ICPEH 2024)*, Physical Research Laboratory (PRL), Ahmedabad, India..
- 2019–2026 **Senior Research Fellowship (SRF), Junior Research Fellowship (JRF), and Master’s Fellowship**, *Astronomical Instrumentation* — awarded by the Indian Institute of Astrophysics (IIA) and Calcutta University..
- 2019 **GATE (Physics)**, *Graduate Aptitude Test in Engineering* — National-level qualification..
- 2018 **CSIR-NET (Physical Sciences)**, *National-level qualification for research fellowship in India*..
- 2013 **COMED-K and AP-EAMCET**, *Merit-based national and state-level engineering entrance examinations*..
- 2011–2017 **Rural Development Trust Scholarship Test (RDTST)**, *Awarded a six-year tuition and hostel fee waiver covering intermediate (10+2) and B.Tech education*..

Conference Presentations

Oral Presentations

- 2026 “**Exoplanets Transmission Spectroscopy Using 2-m Himalayan Chandhra Telescope**”, *Exploring Exoplanets: The Journey to Discover New Frontiers and Habitable Worlds*, International Center for Theoretical Sciences (ICTS), Bengaluru, India.
- 2024 “**Slit and Slitless Low-Resolution Transmission Spectroscopy using the 2m Himalayan Chandra Telescope**”, *30th Young Scientists Conference*, Kyiv, Ukraine.
Virtual presentation
- 2024 “**High- and Low-Resolution Transmission Spectroscopy using the 2m Himalayan Chandra Telescope**”, *42nd Meeting of the Astronomical Society of India (ASI)*, Indian Institute of Science (IISc), Bengaluru, India.
- 2023 “**Opto-mechanical Design of the Multi-object Spectroscopic Mode of the High-Resolution Spectrograph of the Thirty Meter Telescope**”, *Modern Engineering Trends in Astronomy*, Raman Research Institute (RRI), Bengaluru, India.

Poster Presentations

- 2025 “**Atmospheric Dispersion Measurement at Hanle Site**”, *S²-STEP & 6th Indian Planetary Science Conference (IPSC)*, IIT Roorkee, India.
- 2025 “**Simulation of Multi-Object High-Resolution Transmission Spectroscopy**”, *Astronomical Society of India Meeting*, NIT Rourkela, India.
- 2024 “**Transmission Spectroscopy using the 2m Himalayan Chandra Telescope**”, *International Conference on Planets, Exoplanets and Habitability (ICPEH)*, Physical Research Laboratory (PRL), Ahmedabad, India.
- 2023 “**Feasibility of Transmission Spectroscopy using the Hanle Echelle Spectrograph**”, *Strange New Worlds: The Exploration of Exoplanets Conference*, IISER Pune, India.

- 2023 **“Optical Design of Atmospheric Dispersion Corrector for TMT-HROS”**, *41st Meeting of the Astronomical Society of India (ASI), IIT Indore, India.*
- 2023 **“Optical Design of Atmospheric Dispersion Corrector for TMT-HROS”**, *Indo-French CEFIPRA Astronomy Meeting (IFCAM-III), Indian Institute of Astrophysics (IIA), Bengaluru, India.*
- 2024 **“A Multi-functional Fiber Positioning System for Extremely Large Telescopes”**, *SPIE Astronomical Instrumentation Conference, Yokohama, Japan.*
Poster presented by co-author
- 2023 **“Atmospheric Dispersion Corrector for a Multi-object Spectroscopic Mode of HROS-TMT”**, *SPIE Optics and Photonics Conference, San Diego, USA.*
Poster presented by co-author

Telescope Time Allotted

- 1 : 35 full night observation at 2m Himalayan Chandra Telescope, Hanle (India) from 2022 to 2026 as PI.
- 2 : 5 full night observation at Vainu Bappu Telescope, Hanle (India) from 2022 to 2026 as Co-PI.

Workshops and Training

- 2024 **Sagan Exoplanet Summer Hybrid Workshop: Characterizing Exoplanet Atmospheres – The Next Twenty Years**, *NASA Exoplanet Science Institute.*
Virtual participation
- 2024 **Two-HORSEs Conference**, *Max Planck Institute, Germany.*
Virtual participation
- 2024 **AstroAI Workshop**, *Center for Astrophysics | Harvard & Smithsonian, USA.*
Virtual participation
- 2024 **UV-Optical-IR Astronomy in India: Prospects of the Next Decade**, *42nd Meeting of the Astronomical Society of India, IISc Bengaluru, India.*
- 2023 **Machine Learning in Astronomy Workshop**, *41st Meeting of the Astronomical Society of India, IIT Indore, India.*
- 2023 **Sagan Exoplanet Summer Hybrid Workshop: Characterizing Exoplanet Atmospheres – The Next Twenty Years**, *NASA Exoplanet Science Institute.*
Virtual participation
- 2023 **Exoplanets: A Short Course on Planets Orbiting Stars Other Than the Sun**, *Pune Knowledge Cluster (PKC), India.*

Internships

- 2022 **Kodaikanal Solar Observatory**, *Indian Institute of Astrophysics, Kodaikanal, Tamil Nadu, India.*
Research internship
- 2021 **Vainu Bappu Observatory**, *Indian Institute of Astrophysics, Kavalur, Tamil Nadu, India.*
Research internship
- 2020 **Optics Laboratory (Remote Internship)**, *Indian Institute of Astrophysics, Bengaluru, Karnataka, India.*
Remote participation
- 2017 **Diesel Loco Shed Internship**, *Indian Railways, Guntakal, Andhra Pradesh, India.*
Industrial internship

Outreach Activities

- 2026 **Mentor, IIA Job Shadowing Program**, *Indian Institute of Astrophysics*.
Mentored three students as part of the institute outreach program.
- 2026 **Co-mentor for Internship Students**, *Indian Institute of Astrophysics*.
Co-mentored three internship students.
- 2024–2025 **Student Representative, Bhaskara PhD Hostel**, *Indian Institute of Astrophysics*.
Represented a hostel community of about 60 PhD students.
- 2022 **Representative, SCOPE–IIA Outreach Program**, *International Cultural Jamboree, Mudbidri, India*.
Represented IIA outreach initiatives.
- 2021 **Representative, SCOPE–IIA Outreach Program**, *India International Science Festival, Goa, India*.
Represented the Indian Institute of Astrophysics.
- 2021 **Mentor, AISA Camp School**, *Hebbal Community Outreach Program*.
Mentored school students as part of the outreach initiative.
- Participation in Comet Observation Outreach Programs**, *Indian Institute of Astrophysics*.
Public outreach observing activities.
- Contributor**, *Doot Magazine and EdurVed Magazine*, *Indian Institute of Astrophysics*.
Science outreach articles.

References

- Referee 1 **Sivarani Thirupathi**
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- Referee 2 **Ravinder Kumar Banyal**
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- Referee 3 **Shyama Narendranath**
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Research Organisation (ISRO), Bengaluru, India
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